Trading Foundation Architecture Validation: Research Report

Executive Summary

This report evaluates the architectural decisions for a production-grade algorithmic trading research foundation designed for a solo trader on a Windows laptop, targeting BTC 5-minute perpetual futures with LightGBM walk-forward ML. The system—built primarily by Claude Code with zero session memory—requires DuckDB + Parquet for storage, Pydantic v2 + TOML for configuration, a custom feature store with causality testing, walk-forward validation with embargo, and structural enforcement against backtesting errors. Each of the ten design questions is addressed below with current evidence, practitioner consensus, and specific citations.

Question 1: DuckDB for Financial Time-Series Analytics

Adoption in Quantitative Finance

DuckDB has experienced explosive growth, hitting 30,000 GitHub stars by June 2025 and climbing from 1.4% to 3.3% adoption in the Stack Overflow Developer Survey, ranking #4 among databases. In October 2025, DuckDB's in-memory variant reached #1 on ClickBench, the popular analytical database benchmark. The DuckDB 1.4 LTS release (October 2025) added AES-256 encryption, the MERGE statement, Iceberg writes, and significant performance improvements. MotherDuck raised $100M at a $400M valuation to build cloud infrastructure around DuckDB.[1]

While no public evidence points to major hedge funds publishing DuckDB adoption specifically, the broader data engineering

community—including financial analytics pipelines—has embraced it. DuckDB's ability to query Parquet files directly without ETL makes it a natural fit for solo quantitative researchers.[2]

DuckDB vs Polars vs Pandas Performance

Framework	33M Row Benchmark	558K Row Filtering	Key Strength
DuckDB	Fastest[3]	Moderate (optimizable)[4]	SQL interface, Parquet-native, zero-copy Arrow
Polars	Second (lazy eval)[3]	Fast (0.53s)[4]	Rust backend, lazy evaluation
Pandas	Out of memory[3]	Slowest (37.5s)[4]	Ecosystem maturity, familiarity

At the 650K-row, 600-column scale of the described dataset, DuckDB and Polars both perform well. DuckDB's advantage is its SQL interface and direct Parquet querying, while Polars excels with its lazy evaluation and Rust backend. All three frameworks support Apache Arrow, enabling zero-copy conversion between them.[3]

For a financial research workload at this scale, DuckDB as an analytical overlay on Parquet is well-supported. The combination allows SQL-based feature exploration while maintaining Parquet as the canonical storage format.

DuckDB + Parquet for ML Feature Stores

Published architectures combining DuckDB with Parquet for feature stores exist, most notably through Feast. DuckDB serves as a lightweight embedded offline store for historical data, querying Parquet files directly and efficiently. Feast uses `event_timestamp` fields to ensure point-in-time correctness when retrieving features, which is critical for preventing lookahead bias. Feature versioning in this architecture is typically handled through Feast's registry metadata combined with Git-tracked feature definitions.[5][6][7]

Streaming / Real-Time Limitations

DuckDB does not offer a standalone streaming ingestion API. It is fundamentally designed for OLAP workloads with fast columnar processing. However, it can integrate with streaming frameworks through its Python bindings—for example, ingesting from Kafka via batch inserts wrapped in transactions. DuckDB has demonstrated the ability to handle over one million rows inserted per second in streaming patterns. For the transition to live trading, the practical approach is to use DuckDB for historical analysis and a lightweight append mechanism (e.g., periodic Parquet file writes) for incoming data, rather than treating DuckDB as a streaming database.[8][9][10]

Alternatives Assessment

For a single-user Windows financial research system, DuckDB + Parquet is the strongest choice. Polars is a viable alternative for the DataFrame computation layer, and the two can be used together via Arrow interoperability. SQLite is too slow for analytical workloads at this scale. No better-suited embedded analytical database currently exists for this use case.[4][3]

Question 2: Pydantic for ML Pipeline Configuration

ML Experiment Configuration Management (2025–2026)

The ML ecosystem has converged on a pattern of **config files** + **validation** + **experiment tracking**. Hydra (by Meta Research) handles hierarchical configuration by composition and command-line overrides. Weights & Biases and MLflow provide experiment tracking and logging, but neither provides strong config validation at the schema level. The gap that Pydantic fills—runtime type validation with precise error messages—is not natively provided by any of these tools.[11][12][13]

Pydantic v2 Performance

Pydantic v2 is well-suited for ML pipeline configuration. The Rust-based core delivers approximately 5x faster validation than v1, processing ~130K record validations in ~6 seconds (vs ~30 seconds for v1). The v2.11 release added further 5–15% speedups through schema validator reuse. At the scale of hundreds of Optuna trials, each requiring one TOML validation, Pydantic v2 can validate at ~12K operations/second—meaning even 10,000 trials would add under 1 second of total validation overhead. Performance is not a concern.[14][15][16]

Config-as-Code for Reproducibility

Committing configs to Git is the de facto standard for reproducibility in quantitative finance. Martin Fowler's Continuous Delivery for ML framework emphasizes tracking configuration changes alongside data and model versioning. The pattern of TOML config → Pydantic validation → Git commit provides a clean audit trail for every experiment.[17]

TOML vs YAML vs JSON

Format	Strengths	Weaknesses	Python Ecosystem Status
TOML	Human-readable, no whitespace traps, Python PEP-518 standard[18]	Verbose for deeply nested structures[19]	`tomllib` in stdlib since Python 3.11[20]
YAML	Expressive, widely used in ML (Hydra, k8s)	Implicit typing surprises, whitespace sensitivity, complex spec[19]	Requires pyyaml
JSON	Universal, strict parsing	No comments, poor human editability	Built-in `json` module

TOML is the strongest choice for this use case. Python selected TOML via PEP-518 for its package configuration standard. TOML avoids YAML's implicit typing issues (where `3.11` becomes a float instead of a string) and is simpler to hand-edit than JSON. The `tomllib` module is available in Python's standard library from version 3.11 onwards.[18][20][19]

Pydantic-Based ML Pipeline Patterns

The emerging pattern is: TOML/YAML config file → Pydantic model for validation → experiment tracker (MLflow/W&B) for logging. Pydantic's Real Python integration with TOML demonstrates using structural pattern matching and Pydantic validators together for configuration validation. For the described architecture, Pydantic v2 models that map 1:1 to TOML config sections provide strong type safety with negligible performance overhead.[20]

Question 3: Feature Store Patterns for Solo Researchers

Small-Scale Feature Store Patterns

Enterprise feature stores (Feast, Tecton, Hopsworks) are overengineered for a single-user local system. The recommended pattern for solo researchers is a **directory-based feature registry**:

- Each feature family is a Python module implementing a protocol with `compute()`, `causality_test()`, and `output_columns`
- Features are registered via decorators and stored as Parquet files partitioned by computation date
- A lightweight registry (a JSON or TOML file tracked in Git) maps feature names to their module paths, version hashes, and output column schemas[7]

This approach mirrors what academic researchers do in practice—feature computation scripts tracked in version control, with outputs stored as flat files.[7]

Automated Lookahead Bias Detection

Detecting lookahead bias remains fundamentally difficult. As one practitioner on r/quant noted: "What separates a lookahead-biased feature from a highly predictive one?" in a pure statistical test. The state of the art includes:[21]

- **Point-in-time data systems**: Using bitemporal data (knowledge time vs event time) to ensure only historically available data is used[22]
- **Event-driven backtesting**: Processing data sequentially to prevent accidental future-data access[22]
- **Look-Ahead-Bench (January 2026)**: A new standardized benchmark for measuring lookahead bias in financial LLMs, using alpha decay across temporally distinct market regimes to distinguish genuine prediction from memorization[23]
- **Feature shift testing**: Shifting a feature forward in time and measuring performance impact, though this has known limitations[21]

The described system's approach—mandatory `causality_test()` per feature family that structurally prevents future data access—is the right architectural pattern. No off-the-shelf Python library provides built-in causality guarantees for financial features.

tsfresh and Financial Feature Engineering

tsfresh automates time-series feature extraction (autocorrelation, entropy, complexity measures) with built-in relevance testing via the Benjamini-Hochberg procedure. However, tsfresh does not provide causality guarantees—it extracts features from provided windows without enforcing that those windows respect temporal boundaries. It can be used as a feature generation tool within a causality-enforced pipeline, but it does not replace the need for structural lookahead prevention.[24][25]

Feature Versioning

The standard approach is **recomputation from scratch when feature logic changes**, with incremental computation only for appending new time periods to existing features. Feature definitions should be versioned via Git commits, with the registry linking each feature version to its commit hash. Content-based hashing (hashing the feature computation code) can automatically detect when recomputation is needed.[17][7]

Question 4: Walk-Forward Validation Best Practices (2024–2026)

Expanding vs Rolling Window

The current consensus favors **a hybrid approach** rather than dogmatic adherence to either expanding or rolling windows. Empirical findings across 1,455 assets show that optimal window sizes cluster in the 600–800 day (~3-year) range for daily data, and that simple models with daily rolling windows often achieve lower prediction error than complex ML models with static windows. For financial ML, rolling windows are generally preferred when regime changes are expected, as they naturally down-weight stale data.[26][27]

Lopez de Prado's AFML (2018) recommends purged k-fold cross-validation with embargo as the primary validation method, treating walk-forward as one implementation pattern. His core message is that "the hardest problem is not prediction, it's validation". The recommendation has not fundamentally changed, but practitioners increasingly favor rolling windows for assets with demonstrable regime changes.[28][29]

BTC-Specific Considerations

The BTC ETF approval in January 2024 created a documented structural break. Research using Chow tests, rolling correlations, and DCC-GARCH models confirms that BTC's correlation with the S&P 500 increased significantly post-ETF, indicating stronger alignment with equities. The rolling correlation between BTC and equities shifted

from volatile and trending downward to a sharp upward trend immediately after January 2024.[30]

However, by late 2025, BTC's correlation with equities hit yearly lows, dropping to -0.299 with the S&P 500 and -0.24 with the Nasdaq. This suggests that the post-ETF regime may itself be evolving.[31]

Recommendation: Using expanding window with data from 2020–2022 to predict 2025 behavior is questionable given the documented structural break. A rolling window of 12–18 months, or an expanding window with a hard start date no earlier than Q1 2024, better reflects the current market structure.

Embargo Length

The 288-bar embargo (24 hours at 5-minute bars) for a 48-bar (4-hour) label horizon is conservatively appropriate. Published walk-forward implementations use a 30-day embargo for daily data, which is roughly 6x the typical label horizon. The 6:1 ratio in the described system (24h embargo vs 4h label) is within reasonable bounds. Reducing to 144 bars (12 hours, a 3:1 ratio) would increase usable data without significant leakage risk, but the conservative choice is defensible for initial validation.[32]

Minimum Trades Per Fold

No widely cited published benchmark exists for minimum trades per fold at the 5-minute timeframe specifically. However, general statistical guidance suggests a minimum of 30–50 independent trades per fold for basic significance testing, and 200+ trades for reliable distribution estimation. At 5-minute bars with a 4-hour label horizon, the number of independent (non-overlapping) signals per day is limited to approximately 6, meaning each fold needs a minimum of ~5–8 trading days of independent signals for basic reliability, and ~33+ days for robust estimation.

Advanced Fold Selection

Methods beyond fixed-size windows include:

- **Regime-based splitting**: Using hidden Markov models or volatility clustering to define fold boundaries at regime

transitions
- **Combinatorial purged cross-validation (CPCV)**: From Lopez de Prado's work, generating all possible train/test combinations while maintaining temporal ordering and purging[28]
- **Monte Carlo cross-validation**: Random sub-sampling with temporal constraints, as used in recent financial ML research[32]

Question 5: Structural Enforcement in ML Research Workflows

Professional Quant Research Discipline

Lopez de Prado's "10 Reasons Most Machine Learning Funds Fail" (published via GARP) identifies the most common errors: learning side and size simultaneously, using non-unique features, failing to apply fractional differentiation, and not accounting for multiple testing. His Third Law states: "Every backtest must be reported with all trials involved in its production".[33][28]

Professional teams enforce discipline through:

- **Mandatory code review for strategy changes** with dedicated quant validators
- **Automated pipeline gates** that prevent deployment without passing statistical tests
- **Central experiment registries** that log all trials (not just successful ones)[34]
- **Separation of research and production environments** with controlled promotion gates

Defensive ML Engineering

The concept aligns with "defensive programming" in software engineering, applied to ML research. The most common ways validation gets bypassed include:[35][36][22]

1. **Incremental leakage**: Small modifications that individually seem harmless but collectively introduce future information

2. **Survivorship bias**: Testing only on assets that survived the full period
3. **Data snooping through feature selection**: Using the test set (even indirectly) to guide feature choices
4. **P-hacking through sequential testing**: Running many experiments but only reporting the best results
5. **Overfitting to specific time periods**: Strategies tailored to bull or bear markets that fail when regimes change

Holdout Contamination

The standard practice is a **one-time ceremony** for holdout access, but more sophisticated approaches exist. Lopez de Prado's Deflated Sharpe Ratio provides a mathematical framework for controlling multiple testing bias even when holdout data has been accessed multiple times. The key insight is recording the number of independent trials conducted, then adjusting the significance threshold accordingly.[37][34]

The described system's approach—exception-based holdout access requiring explicit ceremony—is sound. Combining this with DSR computation that tracks total trial count provides a quantitative safeguard against cumulative contamination.

Python Tools for Financial ML Discipline

No single Python framework exists specifically for enforcing financial ML research discipline. The closest tools are:

- **MLflow** with mandatory logging (tracks experiments but doesn't enforce gates)[11]
- **Backtrader/Zipline** (event-driven backtesting that prevents some forms of lookahead)
- **Hudson & Thames' mlfinlab** (implements AFML concepts including purged cross-validation)

The described custom enforcement layer (feature registration with causality tests, validated TOML configs, exception-based holdout, mandatory statistical gates) fills a genuine gap in the ecosystem.

Question 6: AI-Assisted Development Knowledge Management

Patterns for Zero-Memory AI Development

The challenge of Claude Code's zero session memory is well-documented. The primary pattern that has emerged is **layered markdown documentation**:[38]

- **CLAUDE.md** in the repo root: Project-level context, architectural decisions, coding conventions (shared with team via Git)
- **CLAUDE.local.md**: Developer-specific notes, excluded from version control
- **Subdirectory CLAUDE.md**: Component-specific context, loaded only when working in that directory
- **~/.claude/CLAUDE.md**: Global cross-project memory merged into every session[38]

Claude Code recursively searches upward from the current working directory to load all relevant CLAUDE.md files. Practitioners report approximately 60% reduction in time re-explaining concepts, with code aligning with project conventions ~85% of the time compared to ~30% without memory.[39][38]

Structured vs Prose for LLM Context

Tables and structured formats are generally more effective than prose for providing LLM context because they minimize token usage while maximizing information density. The described KNOWLEDGE.md (evidence base) and DECISIONS.md (architectural log) pattern aligns with best practices. Key recommendations:

- Use bullet points and structured tables over paragraphs
- Include specific file paths, function names, and code snippets rather than descriptions
- Maintain a "current state" section at the top of STATUS.md that can be quickly scanned[38]

Context Window Degradation

LLM performance consistently degrades as input length increases. Specific findings:[40]

- GPT-4.1 performs well up to ~300K characters (~100K tokens), with dramatic performance drops beyond that threshold[41]
- Claude models show saturation around 16K–32K tokens for retrieval accuracy, with further context offering diminishing returns[42]
- Beyond the practical limit, response times can increase 50x and quality degrades significantly[41]

Recommendation: Keep CLAUDE.md under 5,000 tokens, STATUS.md under 2,000 tokens, and load additional context (KNOWLEDGE.md, DECISIONS.md) only when relevant to the current task. Total context should ideally stay under 80% of the practical limit.[41]

AI-Developer-First Tools

The space is evolving rapidly. Memory MCP servers (like memory-mcp) provide persistent memory through file-based systems with confidence scoring and automatic consolidation. GitHub's Claude Code issue tracker shows persistent memory is among the most requested features. The described four-file system (CLAUDE.md, STATUS.md, KNOWLEDGE.md, DECISIONS.md) is a pragmatic solution that works within current tool constraints.[43][44]

Question 7: Multi-Asset Extension (BTC → NQ/ES)

BTC–NQ/ES Correlation

BTC's correlation with equities is dynamic and has undergone structural changes. Post-ETF approval (January 2024), research using DCC-GARCH models confirmed that BTC's correlation with the S&P 500 increased significantly, with institutional investors treating BTC as a "risk-on" asset similar to technology stocks. However, by late 2025, BTC's short-term correlation with equities dropped to yearly lows (-0.299 with S&P 500, -0.24 with Nasdaq).[45][31][30]

This means the NQ extension is meaningful as an **out-of-distribution test** precisely because the correlation is unstable. A model that works on BTC should not be expected to transfer directly to NQ, but shared features (momentum, volatility clustering, microstructure patterns) can validate whether the feature engineering generalizes.

Feature Engineering Differences

Aspect	BTC Perpetuals	NQ/ES Futures
Trading hours	24/7/365	~23h/day (CME), with gaps
Funding rate	8h (offshore) or 12h (Coinbase)[46]	N/A
Session effects	No open/close auction	Open/close significantly different[47]
Weekend gaps	No gaps	Weekend gaps (Friday close → Sunday open)
Tick size	Variable by exchange	0.25 for ES, 0.25 for NQ
Volume profile	More uniform	Concentrated in RTH hours

Key adaptations needed: session-aware features (RTH vs ETH), gap handling, removal of funding rate features, addition of open interest and settlement-related features.

Data Sources for NQ/ES

For systematic NQ/ES trading, 5-minute bars are appropriate and commonly used. Standard data sources include:

- **CME DataMine**: Official exchange data (tick, 1-min, 5-min)
- **NinjaTrader** + **CQG**: $12/month for all-CME tick data with deep history[48]
- **TradingView**: CFD equivalents (US500, US100) as free proxies for chart analysis[48]
- **Interactive Brokers**: Historical data via API, included with account

Execution Characteristics

For retail position sizes ($10K–$100K), both NQ and ES offer excellent liquidity. E-mini S&P 500 (ES) is the most liquid futures contract in the world. At 5-minute timeframes, slippage of 0.25–0.5 tick per side is typical for ES, and slightly higher for NQ. Commissions of ~$0.85/contract plus ~$1.40 in exchange fees per transaction are standard. BTC perpetuals typically have lower commissions (0.02% per trade on Coinbase) but potentially higher slippage during volatile periods.[47][46]

Regulatory Note

As of July 2025, Coinbase launched CFTC-regulated BTC perpetual futures in the U.S. with up to 10x leverage. This is the first time U.S. retail traders can access perpetual-style futures domestically. However, offshore perpetuals remain generally unavailable to compliant U.S. retail accounts. For NQ/ES, trading through a CFTC-registered broker (e.g., Interactive Brokers, NinjaTrader) is straightforward and fully regulated.[49][50][46]

Question 8: Autonomous ML Experiment Management

Optuna's TPE Sampler for Financial ML

Optuna's TPE (Tree-structured Parzen Estimator) sampler fits Gaussian Mixture Models to partition parameter values into "good" and "bad" groups, choosing parameters that maximize the ratio of the good GMM to the bad GMM. This is well-suited for financial ML optimization because:[51]

- It handles mixed parameter spaces (continuous, discrete, categorical)
- It adapts based on trial history, focusing on promising regions[52]
- MedianPruner can stop poorly performing trials early, saving compute[52]
- TPE became significantly faster in Optuna v4.0.0[51]

No published benchmarks exist specifically for TPE on financial time-series optimization. However, TPE's sample efficiency (requiring fewer trials than random search to find good configurations) makes it practical for walk-forward financial ML where each trial is computationally expensive.

Multiple Testing Correction with Optuna

For the Deflated Sharpe Ratio (DSR), **each Optuna trial counts** toward the total number of trials N. The DSR computes the probability that an estimated Sharpe ratio is statistically significant after controlling for multiple testing, non-normal returns, and sample length.[37][34]

However, Lopez de Prado and Bailey note that the effective number of independent trials may be lower than the raw count when trials are correlated (e.g., TPE's structured search explores similar parameter regions). The key is recording all trials and determining "clusters of effectively independent trials". A conservative approach is to count all Optuna trials as independent, which produces a stricter (more conservative) significance test.[34]

The expected maximum Sharpe ratio after $N >> 1$ independent trials can be approximated analytically, and the DSR adjusts the significance threshold accordingly. A strategy with a high estimated SR that survives DSR correction after, say, 200 trials provides stronger evidence than one that only survives after 10 trials.[37]

Autonomous Research Loop Frameworks

The QuantEvolve framework (2025) demonstrates a multi-agent evolutionary approach to automated strategy discovery: hypothesis generation → strategy implementation → backtest → evaluation → iterative refinement. It uses a feature map aligned with investor preferences to maintain diverse strategies and outperformed conventional baselines across equity and futures markets.[53]

For a simpler autonomous loop, the pattern is:

1. Optuna proposes parameter configurations
2. Walk-forward pipeline trains and evaluates

3. Results are logged with all trial metadata
4. TPE sampler uses history to propose next configuration
5. Human reviews periodically, adjusting search bounds or adding constraints

Meta-Overfitting Detection

Monitoring metrics for detecting when the automated search is overfitting its own search space:

- **Declining out-of-fold performance while in-fold performance improves**: The classic overfitting signal
- **Convergence of trial parameters to narrow ranges**: Indicates the search is exploiting rather than exploring
- **Increasing gap between best trial SR and median trial SR**: Suggests selection bias
- **DSR declining as trial count increases**: Mathematically expected, but a rapid decline suggests the search is finding noise rather than signal[34]

Question 9: Position Sizing for Uncertain Edge

Backtest-to-Live Transition

The consensus recommendation for transitioning from backtest to live trading is aggressive conservatism. Practitioners consistently recommend using **half Kelly or less** as a starting point:[54]

Kelly Fraction	Growth Rate	Volatility	Use Case
Full Kelly	Maximum	Very high	Theoretical only
Half Kelly (f/2)	~75% of full	~50% less volatility[54]	Experienced live traders
Quarter Kelly (f/4)	~50% of full	Much smoother[54]	First live deployment
Eighth Kelly (f/8)	Conservative	Very smooth	Unvalidated backtest edge

One practitioner shared their experience deploying an SPX 0-DTE bot: despite a historical max drawdown of 5%, a sudden regime change caused a 20% loss within weeks of live deployment. This underscores why live performance should be assumed worse than backtest estimates.[55]

Position Sizing for Daily Sharpe 8.77

A daily Sharpe of 8.77 is almost certainly inflated by backtest conditions. For context, an annualized Sharpe of 2.0 is considered excellent in live trading, corresponding to a daily Sharpe of approximately 0.13. A daily Sharpe of 8.77 would imply an annualized Sharpe above 100—virtually impossible in live conditions.

Recommendation for first 6 months of live trading:

1. Start with 1/40 Kelly (as described), which maps to 1–2% risk per trade
2. Further reduce to paper trading or micro-position sizes for the first 30 days
3. Track live-to-backtest performance ratio weekly
4. If live performance sustains >50% of backtest metrics after 60+ trades, gradually increase to 1/20 Kelly
5. Never exceed Quarter Kelly until 6+ months of live data confirms the edge

Adaptive Position Sizing

The self-tuner architecture provides a practical framework: a performance reader queries recent trades, a score computer calculates win rate and rolling metrics, and a multiplier mapper scales position size between floor (0.5x) and ceiling (1.5x) of base size. The critical design decision is lookback window length—too short (5–10 trades) captures variance, too long (50–100 trades) is too slow to detect genuine degradation.[56]

Hard clamps are essential: the system should never size at 0% (missing recovery) or above a safe ceiling (preventing ruin). A circuit breaker separate from the tuner provides a hard stop on consecutive losses.[56]

Question 10: What Am I Missing?

Common Solo Quant Failure Modes

MIT has reported that over 95% of AI pilot applications in corporate settings have failed. For solo quant traders building their own ML systems, the most common failure modes are:[35]

1. **Overfitting masquerading as alpha**: The model learns noise specific to the training period. This is the #1 failure mode[33][35]
2. **Regime blindness**: Models trained on one market regime fail when conditions change[35]
3. **Ignoring market microstructure**: Transaction costs, slippage, and bid-ask spreads can turn a profitable signal into a net loss[35]
4. **Survivorship and lookahead bias**: Often introduced subtly through data handling choices[57][22]
5. **Complexity trap**: Overly complicated systems are harder to debug and more prone to hidden errors[57]
6. **Psychological interference**: Overriding the system during drawdowns, or over-optimizing after a losing streak

Most Likely Missed Validation Step

The described architecture is thorough, but the most commonly missed validation is **realistic transaction cost modeling at the 5-minute timeframe**. At high-frequency intervals, the cost model must account for:

- Maker vs taker fees (which differ significantly for BTC perpetuals)
- Slippage as a function of position size and time-of-day liquidity
- Funding rate payments (for perpetuals held across funding intervals)
- The impact of the bid-ask spread on fill quality

A daily Sharpe of 8.77 that drops to 2.0 or below after realistic costs would indicate the edge is largely an artifact of cost assumptions.

Regulatory Considerations

For a US-based individual running BTC + NQ/ES:

- **BTC perpetual futures**: As of July 2025, CFTC-regulated perpetuals are available via Coinbase Financial Markets (up to 10x leverage). Offshore perpetuals carry enforcement and counterparty risk for US residents[50][49]
- **NQ/ES futures**: Fully regulated via CFTC-registered brokers. Section 1256 contracts receive favorable 60/40 tax treatment (60% long-term, 40% short-term capital gains)[49]
- **Automated trading**: No specific license required for personal account algorithmic trading, but brokers may have their own restrictions on API usage
- **Tax reporting**: All futures trades must be reported on Form 6781 (Section 1256) or Form 8949 (crypto, depending on classification). Crypto tax reporting requirements are evolving[58]

Operational Risks

- **Exchange outages**: Especially critical for BTC perpetuals; funding payments during outages can cause unexpected losses

- **API changes**: Breaking changes in exchange APIs can halt trading; version-pin all API clients
- **Data corruption**: Parquet file corruption can silently produce incorrect features; implement checksums
- **Clock synchronization**: 5-minute bar alignment depends on accurate timestamps
- **Laptop reliability**: A personal laptop is a single point of failure; consider a VPS for live trading
- **Internet connectivity**: Trades cannot be managed during outages; implement dead-man switches

A Professional Quant's First Concern

The most likely first concern upon reviewing this architecture would be the **daily Sharpe of 8.77**. This number is implausibly high for any live strategy and strongly suggests one or more of: incomplete cost modeling, lookahead bias that survived the 10 validation gates, or in-sample optimization that did not fully transfer to out-of-sample evaluation. The recommended response is to treat this number with extreme skepticism, validate costs exhaustively, and expect live performance to be 50–80% worse than backtest estimates. If the system still shows positive expectancy after that haircut, it represents a genuine edge worth deploying conservatively.

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